

ARUNABHA NAG

SDE-1 | FULL-STACK AI DEVELOPER (MERN)

+917998354979 • arunabhadeveloper@gmail.com

SUMMARY

Results-driven Computer Science Engineering undergraduate with hands-on experience designing and deploying production-grade MERN applications, Chrome extensions, and LLM-heuristic pipelines. Proficient in integrating Google Gemini models, optimizing database schemas, and applying OWASP-aligned defensive security protocols. National-level technical hackathons including Smart India Hackathon top-10 and IIT Bombay National Entrepreneurship Challenge AIR 5 (Finalists).

PROJECTS

Verdict Engine | Enterprise Review Analysis & Scoring Pipeline [Full-Stack AI Developer]

TypeScript, React, Chrome Extension, Gemini, Node/Express, MongoDB

- Balanced Scraping Engine:** Built a multi-threaded background scraper that fetches 10 pages in parallel (4 recent, 3 critical, 3 oldest) to pull **100 consolidated reviews** for quality analysis.
- Heuristic-LLM Hybrid Filter:** Implemented a priority scoring system that ranks the 100 scraped reviews, sending only the **top 25 high-priority defect reviews** + 5 general reviews to Gemini (reducing API token costs by **70%**), and backfilling the remaining 70 using rating-heuristic mappings.
- Quantitative Scoring & Math Engine:** Programmed a scoring service calculating authenticity indicators and Tier classifications, reducing full 100+ review analysis latency to **under 3 seconds**.
- Dynamic UI & Edge Cases:** Created a Chrome Extension overlay using React with circular SVG gauges, integrating edge-case states like low-volume banners (1-9 reviews) and insufficient data grey badges (0 reviews).

WealthifyMe | Personal & Shared Finance Tracker [Full-Stack AI Developer]

MERN Stack, Recharts, Stripe, Gemini API, Nodemailer

- AI Financial Assistant:** Integrated the gemini-1.5-flash model to analyze historical transaction records and generate tailored savings plans, automating expense categorization with 85%+ precision.
- Dual-Layer Categorization:** Implemented a live Gemini classification pipeline with a fallback regex engine containing **11 distinct financial rules** (mapping queries to categories like Food, Rent, Transport, and Investments).
- Shared Workspace & Warning Systems:** Built a collaborative household budgeting module supporting multi-user workspace tracking, featuring automated warnings when joint expenses exceed **80%** and **100%** thresholds.
- Stripe Webhook Integration:** Implemented Stripe billing webhooks to securely coordinate tier-progression logic (Free to Pro plan) with zero database state inconsistencies.

Secure MERN Portfolio & Admin Portal [Full-Stack Developer]

MongoDB, Express.js, React.js, Node.js, Tailwind CSS, Helmet.js

- Defensive Security Protocols:** Designed and deployed a secure portfolio site implementing Helmet security headers, IP rate-limiting, and XSS sanitization on form inputs to neutralize NoSQL injections and scripting vulnerabilities.
- OWASP Alignment:** Configured Node/Express backend parser to enforce a strict **10KB body payload limit** to prevent memory exhaustion, and implemented rate-limiting capped at **100 requests per 15 minutes** per IP.
- Dynamic Admin Dashboard:** Constructed a CRUD Admin Portal (/admin) protecting timeline metrics, skills tracks, and inquiry logs with short-lived **2-hour JWT sessions** and bcrypt security question recovery.
- ChunkLoadError Protection:** Implemented a custom React lazy-loading wrapper (lazyWithRetry) that catches ChunkLoadErrors and automatically reloads the browser to pull newly deployed static assets, eliminating blank screen crashes.

WEBSITES AND LINKS

Portfolio Website

<https://arunabhportfolio.vercel.app>

GitHub Link

<https://github.com/Incognitocodernag>

LinkedIn Link

<https://linkedin.com/in/arunabha-nag>

TECHNICAL SKILLS

Languages:

JavaScript, TypeScript, Python, Oracle PL/SQL, MySQL, HTML5, CSS3

Frontend:

React.js, Vite, Tailwind CSS, Recharts, Responsive Web Design

Backend & APIs:

Node.js, Express.js, RESTful APIs, CORS, JWT Session Security

Cloud & DB:

Microsoft Azure, AWS, GCP, MongoDB (Mongoose)

Tools:

Git, GitHub (Gitflow), Power BI, MS Excel

EDUCATION

09/2023 - CURRENT

Srinagar Garhwal, Uttarakhand

Bachelor of Technology - Computer Science and Engineering

School Of Engineering And Technology, H.N.B. Central University (CGPA: 7.0/10.0)

GRADUATED

Dum Dum, Kolkata

Higher Secondary School

Aditya Academy Senior Secondary (CBSE) - Passed with Physics, Chemistry, Maths, Biology with 80% aggregate.

LANGUAGES

ENGLISH	Proficient
HINDI	Proficient
BENGALI	Proficient